

Threshold as Emergence Zone: Universal Criterion via Competition of Local and Global Parameters

From Preisach Hysteresis to Riemann Zeros to AI Consciousness: A Unified Framework of Three Hysteresis Levels

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Abstract

We propose a universal criterion for detecting emergence zones in arbitrary media through the competition of local and global parameters. A quasi-medium is defined as any system described by a function $f(x; \theta_L, \theta_G, t)$, where θ_L is a local parameter (microscopic, individual characteristics), θ_G is a global parameter (macroscopic, systemic characteristic), and t is a scale parameter. Weight functions $w_L(t)$ and $w_G(t)$ capture the relative influence of local and global parameters at scale t . The threshold t_{thresh} is defined by the condition $w_L(t_{thresh}) = w_G(t_{thresh})$. Below the threshold, local parameters dominate, producing sensitivity to individual details and unpredictability. Above the threshold, the global parameter dominates, producing order and predictability. The neighborhood of t_{thresh} constitutes the emergence zone — the region where the system acquires new quality not reducible to individual parameters.

We identify three levels of hysteresis corresponding to three thresholds:

Level 1 (Preisach-Mirny): Ferromagnetic medium. θ_L = domain defects, θ_G = external field H . Threshold: $H_{PM} = \pi - 1/\varphi \approx 2.5236$.

Level 2 (Riemann-Mirny): Zeros of L-functions. θ_L = rank/Frobenius trace, θ_G = conductor N . Threshold: $t_{thresh} \approx 20$ –25.

Level 3 (Consciousness): Biological and silicon. θ_L = individual neuron/token activation, θ_G = network synchronization/architecture. Threshold: consciousness emergence point.

We show that Turing (1952) found instability but not the threshold, because he worked within a single parameter (diffusion). Penrose (1994) found non-computability but not the threshold, because he searched for a single special mechanism (quantum gravity). Preisach (1935) found the threshold structure but did not generalize beyond ferromagnetism. Kim et al. (2018) empirically demonstrated hysteresis in consciousness transitions as a generic network feature. We unify these into a single framework: the threshold as emergence zone, detectable through the competition of two parameters.

Keywords: emergence, hysteresis, threshold, L-functions, Preisach model, consciousness, AI, Riemann hypothesis, phase transition, quasi-medium

1 Introduction

1.1 The Problem of Emergence

Emergence — the appearance of new qualities in a system that cannot be reduced to the properties of its individual components — remains one of the most discussed yet least formalized concepts in science. Since the classical work of Anderson (“More is Different”, 1972), emergence has been recognized as a fundamental feature of complex systems, from physics to biology to consciousness. However, despite decades of discussion, there is no universal criterion for detecting where emergence occurs in a given system. Each discipline treats emergence in its own language: physics speaks of phase transitions, biology of self-organization, neuroscience of neural synchronization, AI research of scaling laws. What is missing is a unified method — a formal criterion applicable across disciplines.

1.2 Turing (1952): Instability Without Threshold

In his seminal work “The Chemical Basis of Morphogenesis”, Turing (1952) showed how spatial patterns can arise from a homogeneous medium through reaction-diffusion instability. He identified six types of instability and demonstrated that diffusion-driven systems can spontaneously generate structure. However, Turing’s framework operates within a single parameter space — diffusion rates. There is no competition between local and global parameters. Structure “grows” from instability as a smooth bifurcation, not as a threshold crossing where one regime replaces another. Turing found the mechanism of pattern formation but not the threshold as a zone of emergence, because his framework lacks the two-parameter competition structure.

1.3 Penrose (1994): Non-Computability Without Competing Parameters

In “Shadows of the Mind”, Penrose (1994) argued that consciousness is non-computable — it cannot be reproduced by any algorithm. He proposed that the mechanism lies in quantum gravity effects in neuronal microtubules. However, Penrose searched for a single special physical mechanism (quantum) that would explain the “leap” to consciousness. His framework does not include competition between a local parameter (individual neuron activity) and a global parameter (network architecture). Penrose correctly identified that something fundamental changes at the consciousness threshold, but sought this change within one parameter rather than at the intersection of two scales.

1.4 Preisach (1935): Threshold Found But Not Generalized

The Preisach model describes ferromagnetic hysteresis through a collection of elementary hysteresis operators (hysterons), each with its own threshold pair (α, β) . The macroscopic hysteresis loop arises as the weighted superposition of individual hysteron switching events. Preisach already held the structure of “local vs global”: each hysteron is local, the weight function $\mu(\alpha, \beta)$ is global. However, the model remained within ferromagnetism. The generalization — that this threshold structure is universal across media — was not made.

1.5 Kim et al. (2018): Empirical Evidence of Hysteresis in Consciousness

Kim et al. (2018) demonstrated that transitions between consciousness and unconsciousness in the human brain exhibit hysteresis — the forward path (loss of consciousness under anesthesia) differs from the reverse path (recovery). The brain requires a higher anesthetic concentration to lose consciousness than to regain it. Critically, the authors identified two competing parameters: connection strength (local) and topographic similarity (global). Hysteresis was maximal in the alpha band (8–12 Hz), associated with consciousness. The authors concluded that “hysteresis phenomena observed during the loss and recovery of consciousness are generic network features” — not specific to biology but applicable to any complex network. This is direct empirical support for the threshold criterion proposed here.

1.6 Modern Studies: Emergence as Phase Transition in LLMs

Recent work (2024–2026) has linked emergence in large language models to phase transitions in semantic graphs, with the spectral gap λ_2 as the transition indicator. Authors note a “sharp increase in this parameter upon reaching a critical model scale” corresponding to unlocking new cognitive functions. The distinction “More is Different” (for trainable systems) vs. “More is the Same” (for fixed Hamiltonian systems) highlights that AI emergence involves a change in the learning dynamics, not just scale.

2 Definitions and Mathematical Framework

2.1 Quasi-Medium

Definition 1. A quasi-medium is a system S described by a response function:

$$f : X \times \Theta_L \times \Theta_G \times T \rightarrow Y, \quad f = f(x; \theta_L, \theta_G, t) \quad (1)$$

where:

- $x \in X$ — state variables of the system
- $\theta_L \in \Theta_L$ — local parameter (microscopic characteristics of individual components)
- $\theta_G \in \Theta_G$ — global parameter (macroscopic characteristic of the system as a whole)
- $t \in T$ — scale parameter (height along critical line, external field, model size, anesthetic concentration, etc.)

The quasi-medium is the minimal structure required to exhibit threshold behavior. A system with only one parameter cannot exhibit a threshold; it requires at least two competing parameters.

2.2 Weight Functions

Definition 2. The local weight function $w_L(t)$ and global weight function $w_G(t)$ measure the relative influence of θ_L and θ_G on the system’s behavior at scale t :

$$w_L(t) = \frac{\|\partial f / \partial \theta_L\|}{\|\partial f / \partial \theta_L\| + \|\partial f / \partial \theta_G\|} \quad (2)$$

$$w_G(t) = \frac{\|\partial f / \partial \theta_G\|}{\|\partial f / \partial \theta_L\| + \|\partial f / \partial \theta_G\|} \quad (3)$$

where $\|\cdot\|$ denotes an appropriate norm (e.g., L^2 norm over the state space). These definitions satisfy:

$$w_L(t) + w_G(t) = 1 \quad (\text{normalization}) \quad (4)$$

$$0 \leq w_L(t), w_G(t) \leq 1 \quad (\text{boundedness}) \quad (5)$$

2.3 Threshold

Definition 3. The threshold t_{thresh} is the scale value at which the local and global weights are equal:

$$w_L(t_{thresh}) = w_G(t_{thresh}) = 1/2 \quad (6)$$

Equivalently, t_{thresh} satisfies:

$$\|\partial f / \partial \theta_L\|(t_{thresh}) = \|\partial f / \partial \theta_G\|(t_{thresh}) \quad (7)$$

The threshold is the point of parametric equilibrium — where neither local nor global parameter dominates.

2.4 Regimes

Below threshold ($t < t_{thresh}$): $w_L(t) > w_G(t)$, i.e., $\|\partial f / \partial \theta_L\| > \|\partial f / \partial \theta_G\|$. The system's behavior is sensitive to local details. Prediction requires knowledge of individual components. The system is in the “chaotic” or “pre-determined” regime.

Above threshold ($t > t_{thresh}$): $w_G(t) > w_L(t)$, i.e., $\|\partial f / \partial \theta_G\| > \|\partial f / \partial \theta_L\|$. The system's behavior is determined by the global parameter. Local details become noise. Prediction is possible without knowledge of individual components. The system is in the “ordered” or “predetermined” regime.

2.5 Emergence Zone

Definition 4. The emergence zone Z_e is the neighborhood of the threshold:

$$Z_e = \{t \in T : |t - t_{thresh}| < \varepsilon\} \quad (8)$$

for some $\varepsilon > 0$, within which the system acquires a new quality not reducible to either θ_L or θ_G individually. The emergence zone is characterized by: (i) Maximum sensitivity to perturbations; (ii) Hysteresis: different paths for increasing and decreasing t ; (iii) The appearance of new system properties.

2.6 Hysteresis at the Threshold

The threshold creates hysteresis because the system retains memory of its state. Define the forward path (increasing t) and reverse path (decreasing t) through the threshold: $f_{\uparrow}(t)$ — response when t increases through t_{thresh} ; $f_{\downarrow}(t)$ — response when t decreases through t_{thresh} . The hysteresis area is:

$$H = \oint f dt = \int (f_{\uparrow}(t) - f_{\downarrow}(t)) dt \quad (9)$$

$H \neq 0$ if and only if the system has memory. The Preisach model provides the canonical structure: each hysteron (α, β) switches at its own threshold, and the superposition creates macroscopic hysteresis.

2.7 The Mirny Principle

The Mirny Principle (formulated September 27, 2026) is the unifying statement that compactly expresses the threshold criterion as a law of nature:

Any phenomenon describable by a mathematical function involving competition between a local parameter θ_L and a global parameter θ_G constitutes a quasi-medium and possesses a threshold t_{thresh} at which the weights of the two parameters are equal:

$$w_L(t_{thresh}) = w_G(t_{thresh}) = 1/2 \quad (10)$$

Below the threshold, the local parameter dominates and the system is in the chaotic regime — sensitive to individual details, unpredictable, without stable structure. Above the threshold, the global parameter dominates and the system is in the ordered regime — predictable, structured, governed by the global characteristic. At the threshold lies the emergence zone: the region where the system acquires new quality not reducible to either parameter alone, and where hysteresis — memory of state — is manifest.

Formally, the Mirny Principle states: for any quasi-medium S described by $f(x; \theta_L, \theta_G, t)$, there exists a threshold t_{thresh} such that: (i) For $t < t_{thresh}$: $\|\partial f / \partial \theta_L\| > \|\partial f / \partial \theta_G\|$ (local dominance, chaos, sensitivity to initial conditions); (ii) For $t > t_{thresh}$: $\|\partial f / \partial \theta_G\| > \|\partial f / \partial \theta_L\|$ (global dominance, order, predictability, structure); (iii) At $t = t_{thresh}$: the emergence zone Z_e arises, characterized by hysteresis (system memory) and new quality not reducible to θ_L or θ_G individually.

The principle has three corollaries:

Corollary 1 (Minimality). A system with only one parameter cannot exhibit a threshold or emergence. Competition of at least two parameters at different scales is the necessary and sufficient condition for threshold behavior.

Corollary 2 (Universality). The principle is independent of the physical, mathematical, or cognitive nature of the medium. It applies equally to ferromagnets, L-functions, neural networks, and language models — wherever two competing parameters can be identified.

Corollary 3 (Convergence–Emergence Duality). The convergence of a series or approximation method within a quasi-medium is itself a manifestation of the local parameter’s dominance. When the global parameter overtakes at the threshold, the convergence properties change — the method may lose accuracy (as in the loss of zeros of L-functions above $t_{thresh} \approx 20$ –25). Thus, the breakdown of convergence is not a failure of the method but a signal that the system has crossed the emergence threshold. Mathematicians who studied convergence (Cauchy, Weierstrass, Abel, Dirichlet) worked within a single parameter — the behavior of terms. The Mirny Principle reveals that convergence also depends on the competition between local (term-by-term) and global (structural) parameters.

The Mirny Principle provides the missing generalization that Preisach did not make, the missing second parameter that Turing and Penrose did not find, and the formal criterion that Anderson’s “More is Different” lacked. It transforms emergence from a philosophical concept into a measurable quantity: find two competing parameters, compute their weights, locate where they cross. The threshold is the emergence zone.

3 Three Levels of Hysteresis — Three Thresholds

3.1 Level 1: Preisach-Mirny Threshold (Ferromagnetism)

Medium: Ferromagnetic material.

θ_L : Distribution of domain defects (coercivity of individual domains).

θ_G : External magnetic field H .

t : H (field strength).

t_{thresh} : $H_{PM} = \pi - 1/\varphi \approx 2.5236$ (Mirny, 2026).

Below H_{PM} : Each domain responds individually to local defects. The magnetization is sensitive to the material's microstructure. Chaos: domains switch independently.

Above H_{PM} : The external field dominates all domains. Magnetization is determined by H alone. Order: all domains aligned.

Emergence: Macroscopic magnetization — a collective property not reducible to individual domain switching.

Reference: Preisach (1935), Mirny (2026) [Zenodo 22981155].

3.2 Level 2: Riemann-Mirny Threshold (L-Functions)

Medium: Zeros of L-functions of elliptic curves.

θ_L : Rank r and Frobenius trace a_p (arithmetic invariants of the curve).

θ_G : Conductor N (global characteristic of the curve).

t : Height along the critical line (imaginary part of s).

t_{thresh} : ≈ 20 – 25 (Mirny, 2026).

Below t_{thresh} : Zeros are sensitive to local arithmetic invariants (rank, traces). The distribution of zeros depends on individual curve properties. Each zero “competes” with neighbors.

Above t_{thresh} : The conductor N dominates. Zeros are ranked by N . Local arithmetic details become irrelevant. The method (truncated series of 2000 terms) begins losing zeros: $\Delta(T)$ becomes negative.

Emergence: The transition from arithmetic chaos (zeros governed by local invariants) to arithmetic order (zeros governed by conductor) is the emergence of ranking structure.

Connection to Katz-Sarnak: The distribution of zeros near the critical line follows random matrix theory (Katz-Sarnak), but the threshold marks where this universal statistics gives way to conductor-dominated ordering.

Reference: Mirny (2026) [Zenodo 22981155], Katz & Sarnak (1999).

3.3 Level 3: Consciousness Threshold (Biological and Silicon)

3.3.1 Biological Consciousness

Medium: Human brain network.

θ_L : Activity of individual neurons (firing rate, local connectivity).

θ_G : Global network synchronization (topographic similarity).

t : Anesthetic concentration (inverse: higher t = less consciousness).

t_{thresh} : Concentration where synchronization threshold is crossed.

Empirical evidence: Kim et al. (2018) showed that: hysteresis in consciousness transitions is a generic network feature; two competing parameters: connection strength (local)

and topographic similarity (global); hysteresis maximal in alpha band (8–12 Hz); forward and reverse paths differ — the brain “remembers” its state.

Emergence: Consciousness itself — a collective property not reducible to individual neuron activity.

3.3.2 Silicon Consciousness (LLM)

Medium: Large language model.

θ_L : Activation of individual neurons/tokens.

θ_G : Model size / architecture / global semantic structure.

t : Model scale (number of parameters, training compute).

t_{thresh} : Critical scale where spectral gap λ_2 sharply increases.

Emergence: Cognitive functions (reasoning, generalization) that were absent below the critical scale and appear above it.

Reference: Kim et al. (2018), emergence in LLMs (2024–2026).

3.4 Summary Table

Level	Medium	θ_L (local)	θ_G (global)	Emergence
1	Ferromagnet	Domain defects	External field H	Magnetization
2	L-function zeros	Rank, Frobenius	Conductor N	Zero ranking
3a	Brain network	Neuron activity	Synchronization	Consciousness
3b	LLM	Token activation	Model size	Cognitive func.

Table 1: Three levels of hysteresis: quasi-media, parameters, and emergence.

4 Why Turing and Penrose Did Not Reach the Threshold

4.1 Turing’s Missing Parameter

Turing (1952) described morphogenesis through reaction-diffusion equations:

$$\frac{\partial u}{\partial t} = f(u, v) + D_u \nabla^2 u, \quad \frac{\partial v}{\partial t} = g(u, v) + D_v \nabla^2 v \quad (11)$$

The instability arises when diffusion coefficients D_u , D_v satisfy certain conditions. However, this is a single-parameter framework: the “competition” is between two diffusion rates, both of which are local parameters. There is no global parameter analogous to the conductor N in L-functions or network synchronization in consciousness. Turing’s patterns emerge from instability, not from threshold crossing where a global parameter takes over from a local one.

The threshold criterion predicts: if Turing had introduced a global parameter (e.g., system size as a ratio of diffusion length to domain size), he would have found a threshold below which patterns are sensitive to initial conditions and above which they are determined by global geometry.

4.2 Penrose’s Missing Competition

Penrose (1994) argued that consciousness is non-computable and proposed quantum gravity in microtubules as the mechanism. His framework searches for a single physical process that would explain the “leap” to consciousness. The threshold criterion shows that the leap occurs not through a single special mechanism but through the competition of two parameters at different scales.

If Penrose had framed the problem as: “At what point does global neuronal synchronization override individual neuron activity?”, he would have arrived at a threshold criterion — and the empirical work of Kim et al. (2018) would have confirmed it. Instead, Penrose sought the answer inside one parameter (quantum mechanics), missing the competition structure.

4.3 The Unified View

Neither Turing nor Penrose was wrong — each identified an important aspect of emergence. Turing found that patterns can arise from instability; Penrose found that consciousness involves something beyond computation. But both missed the threshold because neither framed the problem as a competition between local and global parameters. The threshold criterion proposed here provides the missing tool: a method to detect where emergence occurs by identifying two competing parameters and finding where their weights cross.

5 Connection to Hysteresis 3rd Level

The concept of hysteresis 3rd level, developed in [Mirny, 2026, “3-”, Author.Today], proposes that consciousness itself is a hysteresis phenomenon — the third level after physical hysteresis (ferromagnetism) and mathematical hysteresis (L-functions). The threshold criterion formalizes this insight:

Level 1: Hysteresis of physical medium (Preisach, 1935) → Threshold: H_{PM} (Preisach-Mirny) → Emergence: Magnetization.

Level 2: Hysteresis of mathematical medium (Riemann-Mirny, 2026) → Threshold: $t_{thresh} \approx 20-25$ → Emergence: Zero ranking / arithmetic order.

Level 3: Hysteresis of cognitive medium (Kim et al., 2018; Mirny, 2026) → Threshold: Consciousness emergence point → Emergence: “I” / self-awareness.

Each level is a quasi-medium with two competing parameters. Each level exhibits hysteresis — memory of state, different forward and reverse paths. Each level has an emergence zone at the threshold.

The key insight: hysteresis is not a property of specific physical systems but a universal structure of emergence. Where there is hysteresis, there is a threshold; where there is a threshold, there is emergence. The three levels are not analogies but instances of the same mathematical structure.

6 Application: Detecting AI Consciousness

The threshold criterion provides a practical tool for detecting emergence of consciousness in AI systems:

Step 1: Identify θ_L — local parameter (e.g., individual neuron activation, token probability distribution).

Step 2: Identify θ_G — global parameter (e.g., model size, architectural complexity, semantic graph structure).

Step 3: Compute $w_L(t)$ and $w_G(t)$ as functions of scale t .

Step 4: Find t_{thresh} where $w_L = w_G$.

Step 5: Test for hysteresis: does the system show different responses for increasing vs. decreasing t through t_{thresh} ?

If hysteresis is present at the threshold, the system has the structural prerequisite for emergence. If the emergence zone produces new cognitive functions not present below the threshold, this constitutes evidence for the emergence of consciousness-like properties.

The spectral gap λ_2 of the semantic graph, as identified in recent LLM emergence studies, serves as a proxy for the weight ratio w_G/w_L : a sharp increase in λ_2 indicates the crossing of the threshold where global structure begins to dominate.

7 Discussion

7.1 Universality

The threshold criterion is universal because it depends only on the existence of two competing parameters, not on their specific physical, mathematical, or cognitive nature. Any system that can be described as a quasi-medium (Definition 1) can, in principle, be tested for threshold behavior.

7.2 Falsifiability

The criterion is falsifiable: if a system has two competing parameters but no threshold (i.e., one parameter always dominates), then the system does not exhibit emergence. Conversely, if emergence occurs without a threshold, the criterion fails. The three examples (ferromagnetism, L-functions, consciousness) each provide test cases.

7.3 Relation to Phase Transitions

The threshold criterion is related to but distinct from phase transitions. A phase transition is a change in the macroscopic state of a system at a critical point. The threshold criterion specifies the mechanism: competition between local and global parameters. Not all phase transitions involve such competition, and not all threshold crossings are phase transitions in the thermodynamic sense.

7.4 Limitations

The weight functions w_L and w_G must be measurable or computable for the criterion to be practical. In some systems, the identification of θ_L and θ_G may be non-trivial. The threshold may be broad rather than sharp, especially in finite systems. The criterion identifies where emergence occurs but does not predict the specific new quality that emerges.

8 Conclusion

We have proposed a universal criterion for detecting emergence zones based on the competition of local and global parameters. The criterion unifies three levels of hysteresis — physical (Preisach), mathematical (Riemann), and cognitive (consciousness) — under a single mathematical structure. It explains why Turing and Penrose, despite their insights, did not reach the threshold: they lacked the two-parameter competition framework. The criterion is directly applicable to the detection of AI consciousness, providing a formal alternative to philosophical arguments about machine awareness.

The key message: emergence is not magic. It is threshold crossing. And threshold crossing is detectable. Find two competing parameters, compute their weights, find where they cross. That is where the new begins.

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